```
# Autoencoder for Image Denoising using PyTorch

import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader
from torchvision import datasets, transforms
import matplotlib.pyplot as plt
import numpy as np
from torchsummary import summary
```

```
# Device configuration
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
```

```
# Transform: Normalize and convert to tensor
transform = transforms.Compose([
    transforms.ToTensor()
])
```

```
# Load MNIST dataset
dataset = datasets.MNIST(root='./data', train=True, download=True, transform=transform)
test_dataset = datasets.MNIST(root='./data', train=False, download=True, transform=transform)

train_loader = DataLoader(dataset, batch_size=128, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=128, shuffle=False)
```

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100%|██████████| 9.91M/9.91M [00:01<00:00, 5.82MB/s]
100%|██████████| 28.9k/28.9k [00:00<00:00, 153kB/s]
100%|██████████| 1.65M/1.65M [00:01<00:00, 1.47MB/s]
100%|██████████| 4.54k/4.54k [00:00<00:00, 9.54MB/s]
```

```
# Add noise to images
def add_noise(inputs, noise_factor=0.5):
    noisy = inputs + noise_factor * torch.randn_like(inputs)
    return torch.clamp(noisy, 0., 1.)
```

```
# Define Autoencoder
class DenoisingAutoencoder(nn.Module):
    def __init__(self):
      super(DenoisingAutoencoder,self).__init__()
      self.encoder=nn.Sequential(
          nn.Conv2d(1, 16, 3, stride=2, padding=1),
          nn.ReLU(),
          nn.Conv2d(16, 32, 3, stride=2, padding=1),
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            nn.ReLU()
        )
        self.decoder=nn.Sequential(
            nn.ConvTranspose2d(32, 16, 3, stride=2, padding=1, output_padding=1),
            nn.ReLU(),
            nn.ConvTranspose2d(16, 1, 3, stride=2, padding=1, output_padding=1),
            nn.Sigmoid()
        )
    def forward(self,x):
        x=self.encoder(x)
        x=self.decoder(x)
        return x
```

```
# Initialize model, loss function and optimizer
model = DenoisingAutoencoder().to(device)
criterion = nn.MSELoss()
optimizer = optim.Adam(model.parameters(), lr=0.001)
```

```
# Print model summary
summary(model, input_size=(1, 28, 28))
```

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1           [-1, 16, 14, 14]             160
              ReLU-2           [-1, 16, 14, 14]               0
            Conv2d-3             [-1, 32, 7, 7]           4,640
              ReLU-4             [-1, 32, 7, 7]               0
   ConvTranspose2d-5           [-1, 16, 14, 14]           4,624
              ReLU-6           [-1, 16, 14, 14]               0
   ConvTranspose2d-7            [-1, 1, 28, 28]             145
          Sigmoid-8            [-1, 1, 28, 28]               0
================================================================
Total params: 9,569
Trainable params: 9,569
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.00
Forward/backward pass size (MB): 0.13
Params size (MB): 0.04
Estimated Total Size (MB): 0.17
----------------------------------------------------------------
```

```
# Training Function
def train(model, loader, criterion, optimizer, epochs=5):
    model.train()
    for epoch in range(epochs):
        running_loss = 0.0
        for images, _ in loader:
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            images = images.to(device)
            noisy_images = add_noise(images).to(device)

            outputs = model(noisy_images)
            loss = criterion(outputs, images)

            optimizer.zero_grad()
            loss.backward()
            optimizer.step()

            running_loss += loss.item()
        print(f"Epoch [{epoch+1}/{epochs}], Loss: {running_loss/len(loader):.4f}")
```

```
# Evaluate and visualize
def visualize_denoising(model, loader, num_images=10):
    model.eval()
    with torch.no_grad():
        for images, _ in loader:
            images = images.to(device)
            noisy_images = add_noise(images).to(device)
            outputs = model(noisy_images)
            break

    images = images.cpu().numpy()
    noisy_images = noisy_images.cpu().numpy()
    outputs = outputs.cpu().numpy()

    print("Name:  VUTUKURI SAI KIUMAR REDDY                ")
    print("Register Number:   212224230307          ")
    plt.figure(figsize=(18, 6))
    for i in range(num_images):
        # Original
        ax = plt.subplot(3, num_images, i + 1)
        plt.imshow(images[i].squeeze(), cmap='gray')
        ax.set_title("Original")
        plt.axis("off")

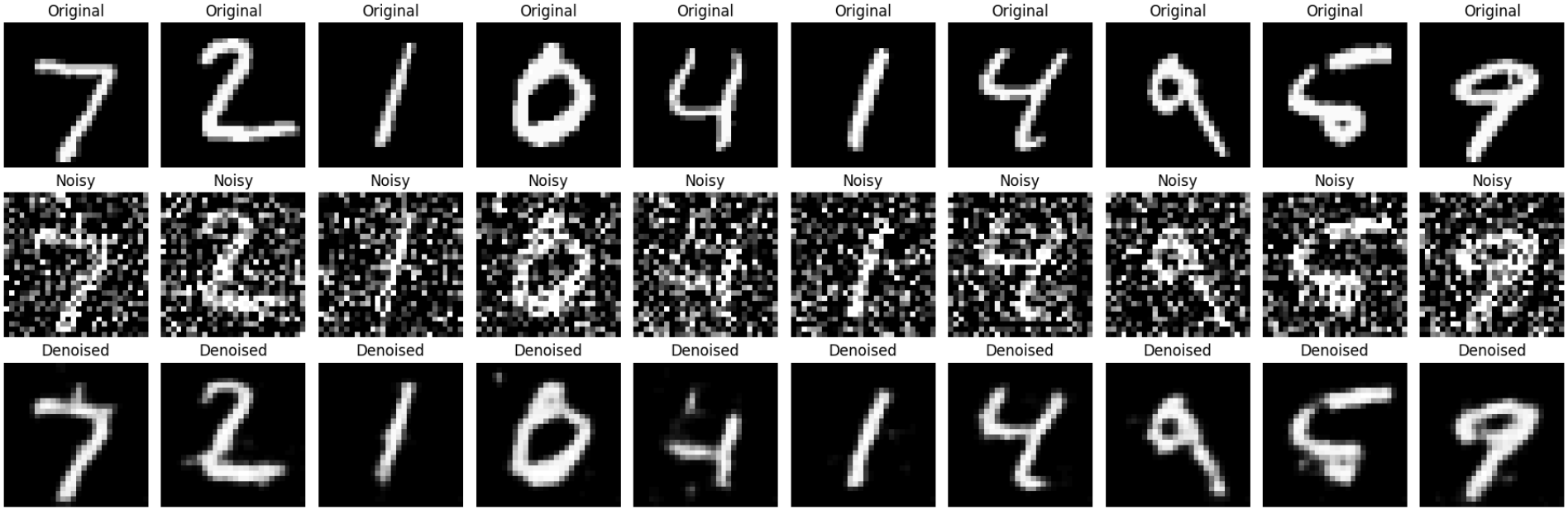
        # Noisy
        ax = plt.subplot(3, num_images, i + 1 + num_images)
        plt.imshow(noisy_images[i].squeeze(), cmap='gray')
        ax.set_title("Noisy")
        plt.axis("off")

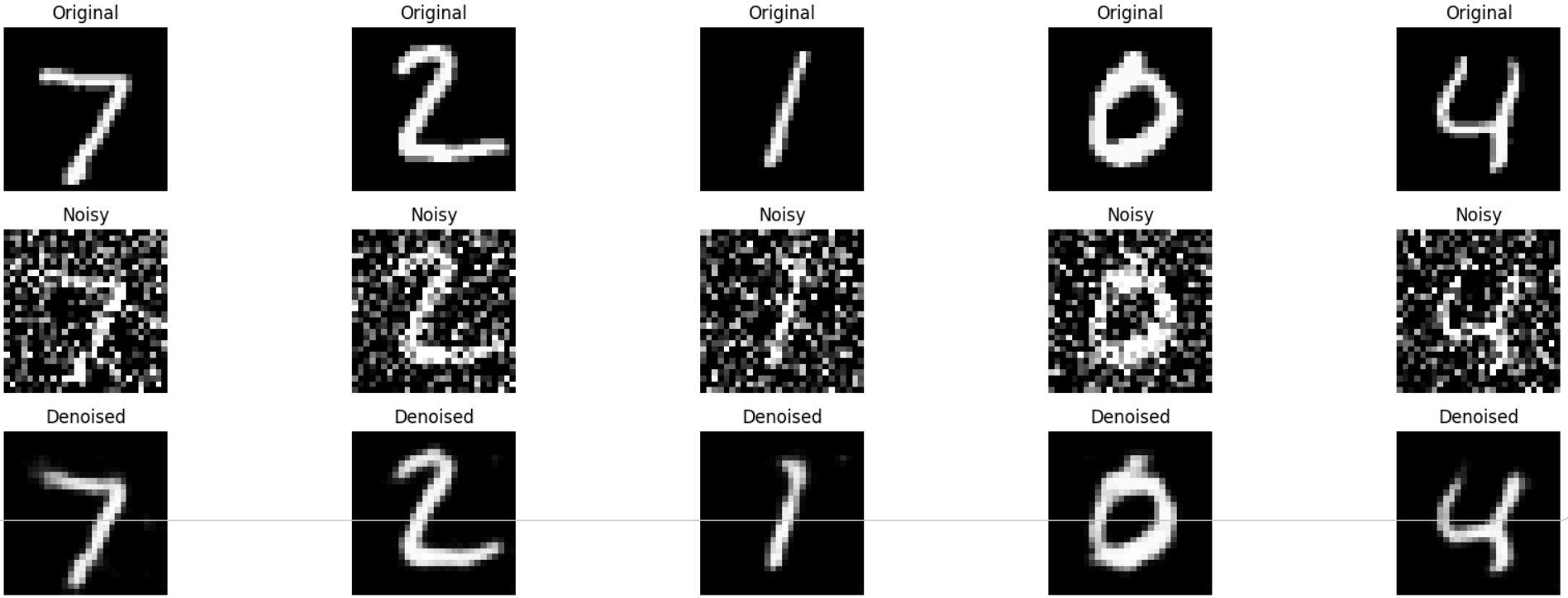
        # Denoised
        ax = plt.subplot(3, num_images, i + 1 + 2 * num_images)
        plt.imshow(outputs[i].squeeze(), cmap='gray')
        ax.set_title("Denoised")
        plt.axis("off")
```

```
    plt.tight_layout()
    plt.show()
```

```
# Run training and visualization
train(model, train_loader, criterion, optimizer, epochs=10)
visualize_denoising(model, test_loader)
```

```
Epoch [1/10], Loss: 0.0673
Epoch [2/10], Loss: 0.0155
Epoch [3/10], Loss: 0.0144
Epoch [4/10], Loss: 0.0139
Epoch [5/10], Loss: 0.0136
Epoch [6/10], Loss: 0.0134
Epoch [7/10], Loss: 0.0133
Epoch [8/10], Loss: 0.0131
Epoch [9/10], Loss: 0.0130
Epoch [10/10], Loss: 0.0129
Name:  VUTUKURI SAI KIUMAR REDDY
Register Number:   212224230307
```




```
visualize_denoising(model, test_loader, num_images=5)
```

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